

Learning Outcomes (relevant to this test)

By this stage, students should be able to:

LO1. Apply Gauss' law to derive electric field expressions for symmetric charge distributions.

LO2. Relate electric field and potential and compute potential differences.

LO3. Explain dielectric behaviour using polarization and boundary conditions.

LO4. Model multi-conductor systems using capacitance matrices.

LO5. Interpret physical meaning of Maxwell's equations (electrostatics & magnetostatics).

Marking Rubric by Question

Question 1 (20 marks)

Assesses: LO1, LO2, LO5

(a)(i) Electric field derivation [5 marks]

Criteria	Marks
Correct application of Gauss' law (chooses cylindrical surface, symmetry argument)	2
Correct expression for enclosed charge and surface area	1
Correct final expression $E = \frac{\rho_l}{2\pi\epsilon_0 r}$	2

Common errors & partial marks

- Uses wrong Gaussian surface → max 2 marks
 - Correct approach but algebra error → deduct 1 mark
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(a)(ii) Potential difference derivation [5 marks]

Criteria	Marks
Correct relationship $V = -\int E dr$	1
Correct substitution of field expression	1
Correct integration limits	1
Logarithmic form obtained	1
Final correct expression	1

(a)(iii) Ground plane assumption [2 marks]

Criteria	Marks
Mentions symmetry breaking due to ground	1
Explains need for image method / altered boundary conditions	1

(b) Numerical potential calculation [5 marks]

Criteria	Marks
Correct identification of r_1 and r_2	1
Correct substitution of values	2
Correct numerical answer (within tolerance)	2

(c) Method of images concept [3 marks]

Criteria	Marks
Recognises ground as conducting boundary	1
Explains image charge idea	1
Links to capacitance calculation	1

Question 2 (16 marks)

Assesses: LO3

(a) Polarization vs Polarization vector [4 marks]

Criteria	Marks
Defines polarization as dipole alignment	1
Defines polarization vector as dipole moment per unit volume	2
Distinguishes process vs vector quantity	1

(b) Polarization vector calculation [8 marks]

Criteria	Marks
Correct electric field calculation	2
Correct relation $\mathbf{P} = \epsilon_0(\epsilon_r - 1)\mathbf{E}$	2
Correct substitution	2
Correct final vector (magnitude + direction)	2

(c) Boundary condition reasoning [4 marks]

Criteria	Marks
States boundary condition $D_{n1} = D_{n2}$	1
Uses $D = \epsilon E$ relationship	1
Shows different permittivity leads to different E	1
Concludes $E_1 \neq E_2$	1

Question 3 (10 marks)

Assesses: LO4

(a) Capacitance matrix formation [8 marks]

Criteria	Marks
Correct matrix structure	2
Correct placement of mutual capacitances	3
Correct diagonal terms (sum of connected capacitances)	2
Units and symmetry noted	1

(b) Advantage of capacitance matrix [2 marks]

Criteria	Marks
Mentions simplification of multi-conductor analysis	1
Mentions relation $Q = CV$ in matrix form	1

Question 4 (4 marks)

Assesses: LO5

Criteria	Marks
States observation: No magnetic monopoles	2
States observation: Magnetic fields produced by currents	2

Performance Levels (for overall grading feedback)

Level	Description
Excellent (HD/D)	Correct reasoning, clear steps, accurate answers, good physical interpretation

Level	Description
Competent (C)	Mostly correct, minor errors, understanding evident
Developing (P)	Partial understanding, some key errors, steps unclear
Needs Improvement (F)	Major misconceptions, incomplete reasoning

Alignment Summary

Question Learning Outcome

Q1	LO1, LO2, LO5
Q2	LO3
Q3	LO4
Q4	LO5

Marking Notes for Tutors

- Award method marks generously where concept is correct.
- Deduct marks for:
 - Missing units (max 1 mark deduction per question)
 - No working shown (max 50% of marks)
- Encourage conceptual explanations in words for full marks.